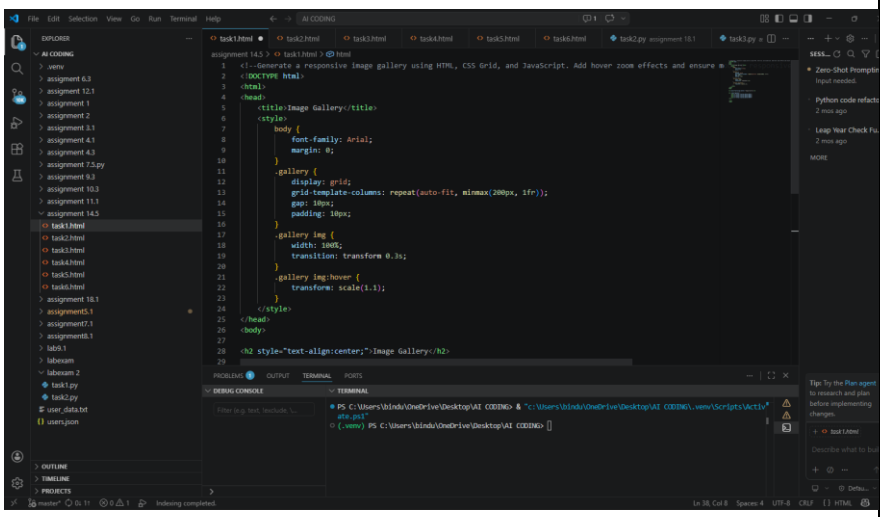
SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
NAME:	**A.Pushpanjali**	**HALLTICKENT NUMBER**	2303A54050
Duration	2 Hours	**Batch**	47_B
Assignment Number: 14.5			

Q.No.	Question	*Expected Time to complete*
1	**Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation** **Lab Objectives** • Design **functional, visually appealing** web applications using HTML, CSS, and JavaScript with AI assistance. • Apply **responsive, accessible, and interactive design principles**. • Create **practical UI components** for real-world web applications. • Use AI to optimize **layout, UX, and performance**. **Task 1 – Image Gallery Website** Create an image gallery web page. **Scenario** Create a responsive image gallery using HTML, CSS Grid, and basic styling. The gallery should adjust to different screen sizes and include hover effects. **Prompt Used** Generate a responsive image gallery using HTML and CSS Grid	Week7 - Friday

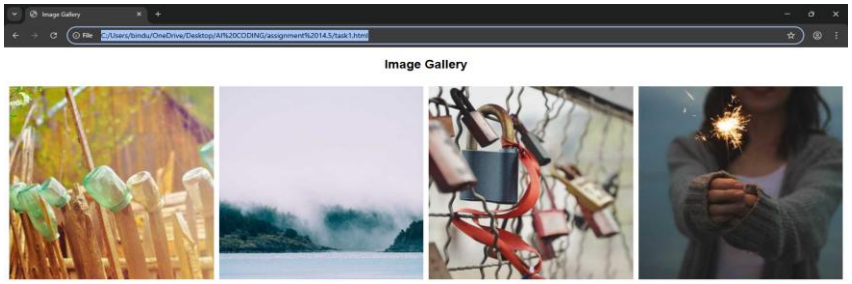
with hover zoom effects and mobile responsiveness.

CODE:



```html
<!--Generate a responsive image gallery using HTML, CSS Grid, and JavaScript. Add hover zoom effects and ensure m
<!DOCTYPE html>
<html>
<head>
    <title>Image Gallery</title>
    <style>
        body {
            font-family: Arial;
            margin: 0;
        }
        .gallery {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 10px;
            padding: 10px;
        }
        .gallery img {
            width: 100%;
            transition: transform 0.3s;
        }
        .gallery img:hover {
            transform: scale(1.1);
        }
    </style>
</head>
<body>

<h2 style="text-align:center;">Image Gallery</h2>
```

OUTPUT :



Explanation

This program creates an image gallery using CSS Grid. The images automatically adjust based on screen size. A hover effect is added to make the images zoom when the user points at them.

Task 2 – Profile Card Generator

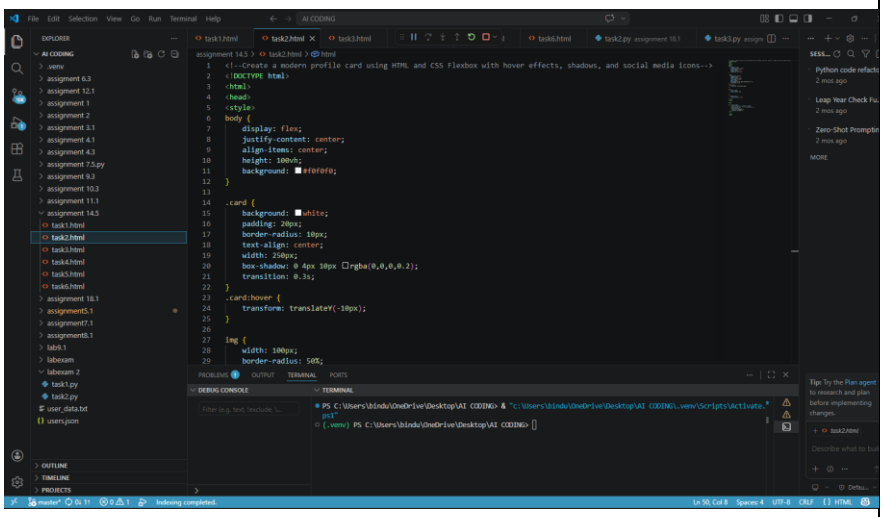
Scenario

Design a profile card using HTML and CSS with hover effects and user details.

Prompt Used

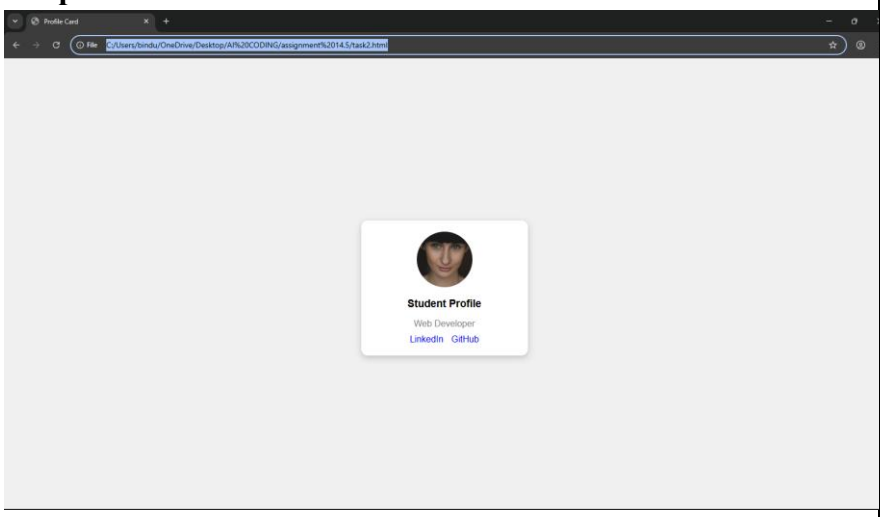
Create a modern profile card using HTML and CSS Flexbox with hover effects and shadow styling.

Code



```html
<!--Create a modern profile card using HTML and CSS Flexbox with hover effects, shadows, and social media icons-->
<!DOCTYPE html>
<html>
<head>
<style>
body {
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    background: #f0f0f0;
}

.card {
    background: white;
    padding: 20px;
    border-radius: 10px;
    text-align: center;
    width: 250px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.2);
    transition: 0.3s;
}
.card:hover {
    transform: translateY(-10px);
}

img {
    width: 100px;
    border-radius: 50%;
```

Output:



Explanation:

This program creates a simple profile card. CSS is used to center the card and add shadow effects to make it look attractive.

Task 3 -Develop a dynamic shopping cart system.

Requirements:

Scenario:

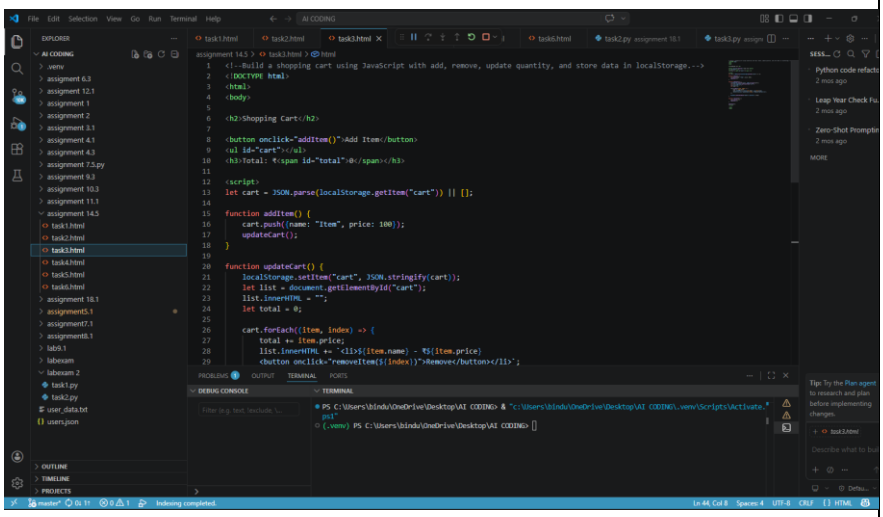
Create a simple shopping cart using JavaScript with add and remove functionality.

Prompt Used:

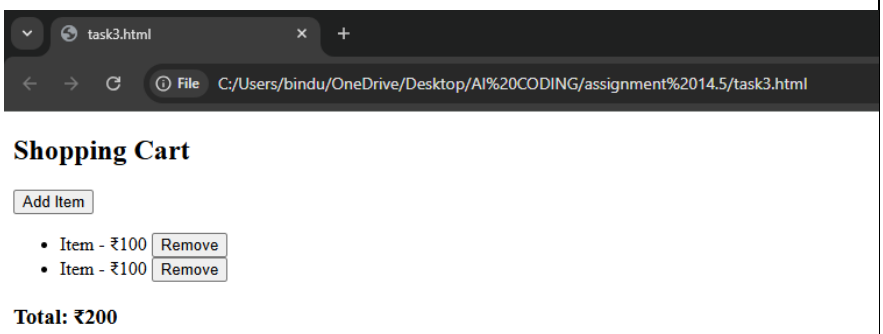
Build a shopping cart using JavaScript with add and remove item

functionality and display total price.

CODE:



OUTPUT:



Explanation

This program uses JavaScript to store items in an array and display them dynamically on the web page.

Task 4 – Role-Based Access Control Application

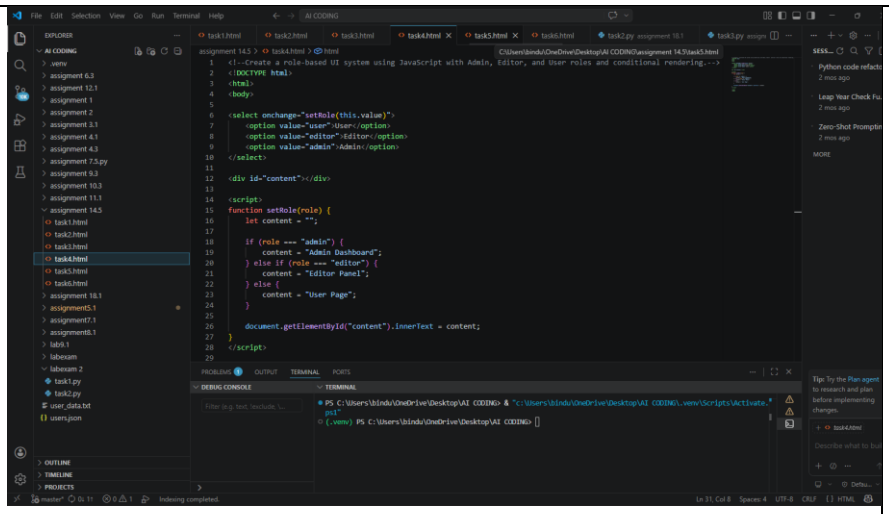
Scenario

Create a system where content changes based on user role (Admin/User).

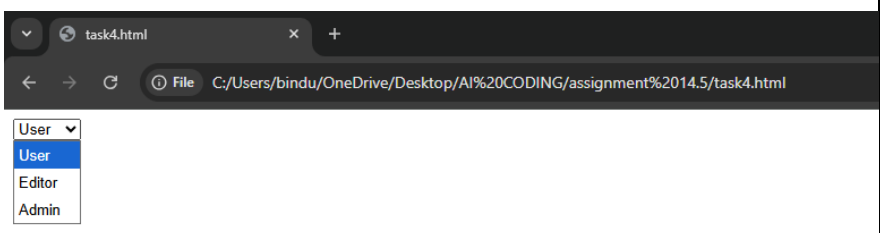
Prompt Used

Create role-based UI using JavaScript with conditional rendering for admin and user roles.

CODE:



OUTPUT:



Explanation

The program checks the selected role and displays content accordingly using conditional statements.

Task 5 – Multi-Language Website

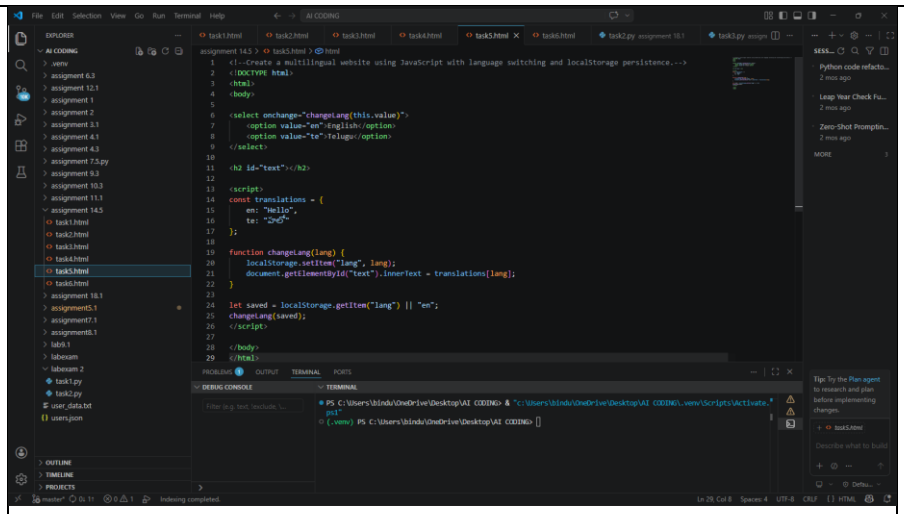
Scenario

Create a website that supports multiple languages.

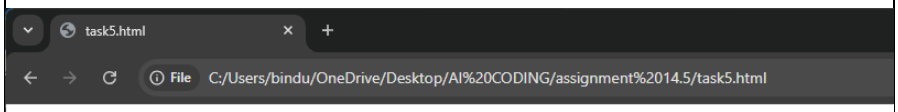
Prompt Used

Create a multilingual website using JavaScript with language switching.

CODE:

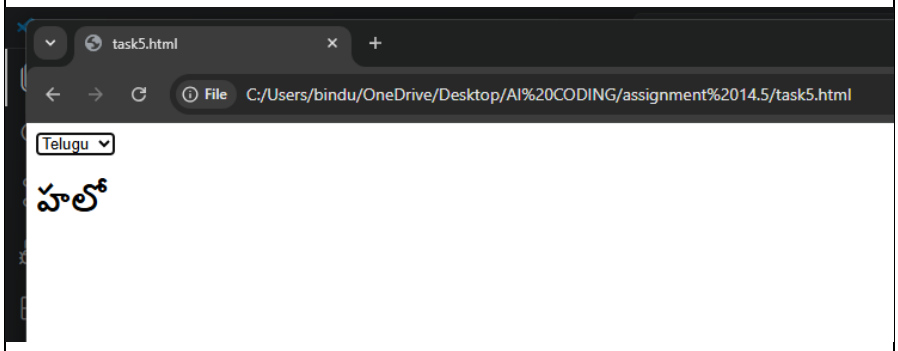


OUTPUT:



English

Hello



Explanation

This program switches text based on selected language using JavaScript conditions.

Task 6 – Online Examination System

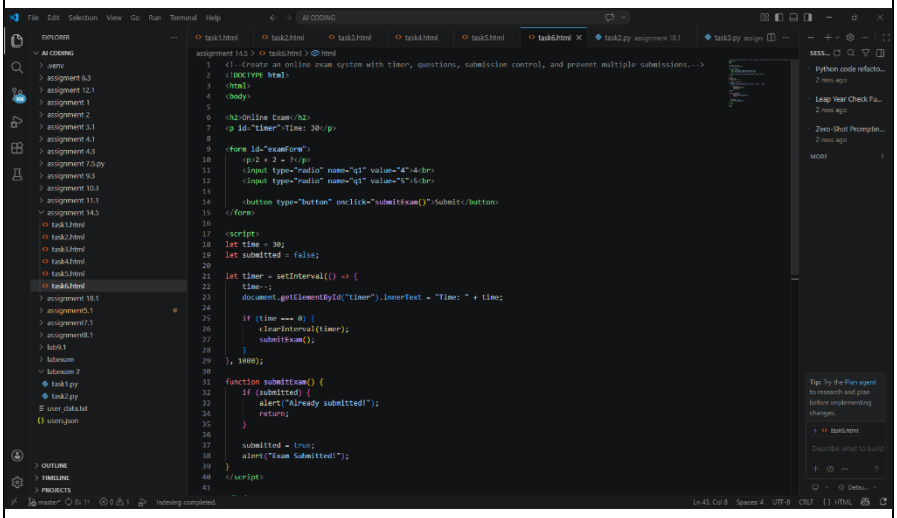
Scenario

Create a simple online exam with a timer.

Prompt Used

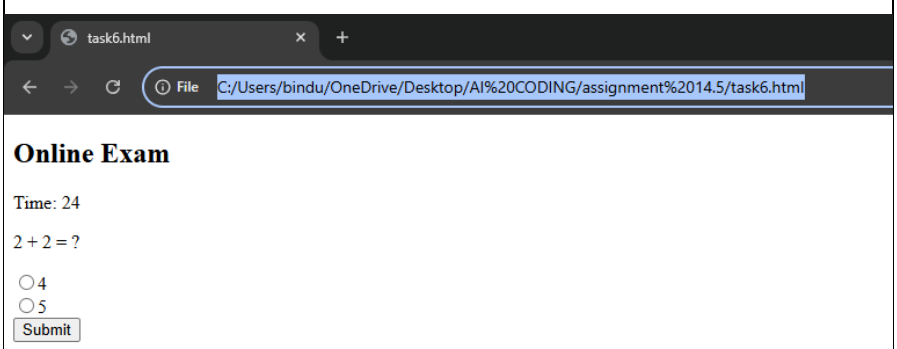
Create an online exam system using JavaScript with timer and submit functionality.

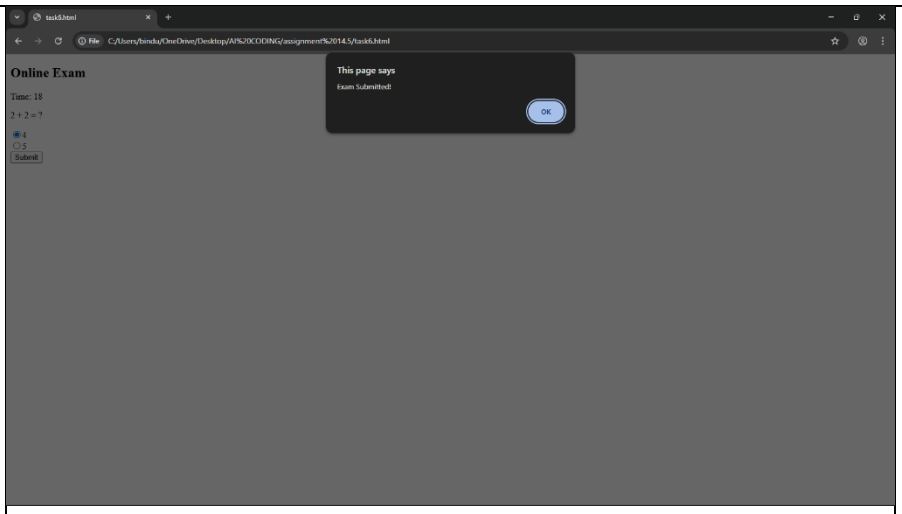
Code



```html
<!--Create an online exam system with timer, questions, submission control, and prevent multiple submissions.-->
<!DOCTYPE html>
<html>
<body>

<h2>Online Exam</h2>
<p id="timer">Time: 30</p>

<form id="examForm">
    <p>2 + 2 = ?</p>
    <input type="radio" name="q1" value="4">4<br>
    <input type="radio" name="q1" value="5">5<br>

    <button type="button" onclick="submitExam()">Submit</button>
</form>

<script>
let time = 30;
let submitted = false;

let timer = setInterval(() => {
    time--;
    document.getElementById("timer").innerText = "Time: " + time;

    if (time === 0) {
        clearInterval(timer);
        submitExam();
    }
}, 1000);

function submitExam() {
    if (submitted) {
        alert("Already submitted!");
        return;
    }

    submitted = true;
    alert("Exam Submitted!");
}
</script>
```

Output





Explanation

This program uses JavaScript timer to decrease time every second and allows submission.